

Quantifying the Effects of Vertical Shielding Due to Lead at Jungfraujoch Cosmic Ray Observatory

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Abstract

The attenuation of cosmic ray flux by lead shielding was investigated using a MiniPIX detector at the Jungfraujoch Observatory. A lead shielding apparatus was incrementally constructed around the detector using standardized bricks, with vertical shielding thickness increased systematically across successive trials. Experimental results demonstrate that two vertical layers of lead shielding were sufficient to produce a statistically significant reduction in electron detection rates, consistent with theoretical predictions for electron attenuation in high density materials. These findings demonstrate the effectiveness of lead as a shielding medium for low to moderate energy cosmic rays.

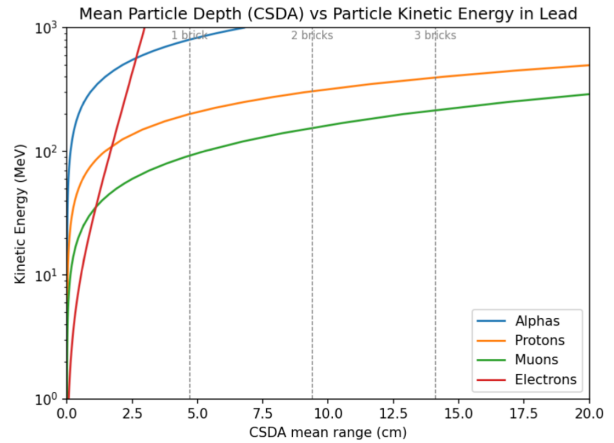


Figure 1: Source from PDG and NIST

1 Introduction

Lead is a common material used for radiative shielding, primarily due to its high density of 11.348 g/cm^3 , high atomic number, and high linear and mass attenuation coefficients [3]. Shielding with lead causes high-energy particles to interact with a large amount of matter (even over a relatively small physical distance), lose energy via Compton scattering and the photoelectric effect, and thereby reduces radiation intensity at a significantly higher rate than other materials with lower attenuation coefficients.

The Bethe-Bloch equation estimates the depth radiation will travel through lead [4][2]. As the above figure shows, muons and protons above 500 MeV are expected to be unaffected by shielding of electrons, and alphas on the order of a few GeV will be stopped quite easily. Radiative stopping in lead is significant for electrons with a critical energy of 7.43 MeV and muons the effects are expected to be far less significant with a critical energy of 141 GeV[2]. With the Muon energy spectrum being nearly 0 below 1 GeV

in atmosphere and the main peak being at 10 GeV only losing on average a few GeV when traveling from the atmosphere, it is not expected that muon shielding or muon radiative effects will be detected by our experiment [1]. For electrons, radiative effects are definitely significant for us, in particular the peak energy dissipation of a cascade from electrons between 2-50 GeV will be roughly 4 cm or a similar width to our bricks [2]. For all alphas and protons, radiative effects may be neglected, as their critical energies are out of the scope of cosmic rays[2].

2 Methods

The shielding properties of lead were investigated by covering the detector with successive layers of lead bricks, with length 14.6 cm, width 7.5 cm, and height 4.7 cm. The experiment was conducted in the library of the observatory, chosen for its increased particle flux and ease of access. All measurements used a MiniPIX detector lent to the team by the experts

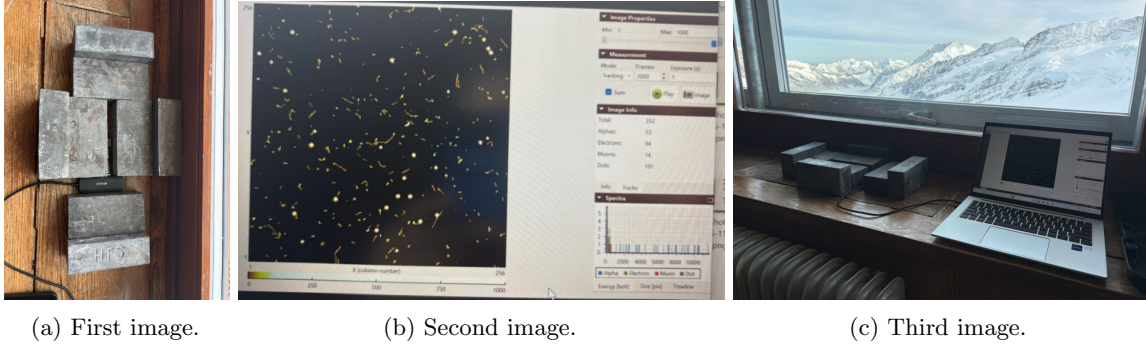


Figure 2: From left to right: A picture of the baseline test of our experiment (as described above), a screenshot of our MiniPIX data after the baseline test was conducted (these screenshots were saved onto a flash drive, and the data contained within was used in our analysis), and a larger-scale view of our experiment in the observatory's library.

at CERN, which identifies particle type to a certain degree in addition to counting events. A single 5-minute baseline measurement without shielding was recorded to first establish ambient flux. Subsequent trials introduced lead shielding in a systematic series.

The team first surrounded the detector with a horizontal plane of lead blocks to restrict observations to particles arriving from directly above. Up to three successive vertical layers of lead, of thickness 4.7 cm, were added above the detector to probe how cumulative material thickness attenuates cosmic ray flux. At each configuration, the team recorded the total particle count and the counts per type as identified by the MiniPIX.

3 Analysis

A statistically significant decrease in particle rates across most particle types was observed as the level of shielding was increased. Of particular interest were electrons, as they were identified in high quantities and are likely easily shielded, and “dots”, high-energy particles the MiniPIX could not identify. These were expected to be a mix of high energy photons and other particles incident orthogonal to the detector plane. As expected, the baseline measurement with zero shielding produced much higher total counts than the measurements with side shielding. The one exception was in alpha particles, which increased significantly. This suggests the lead may have contained small alpha-emitting impurities. To prevent this from skewing the results, alpha counts were omitted from the following analysis. Finally, muons were detected at such a low rate, and experienced such a small amount of attenuation, that they were also not analyzed.

If the dots detected by the MiniPIX were primarily photons, then an exponential decrease in intensity as a function of shielding thickness would be seen. This is because photons have no charge and do not follow the Bethe-Bloch equation. They are thus best modeled with an exponential decay in flux with some characteristic interaction length. Our observed dot counts are shown in Figure 2. Note that the detection of cosmic rays was modeled as a Poisson process. This gave a percent error in the observed rate of $\frac{1}{\sqrt{N}}$, where N is the number of detections.

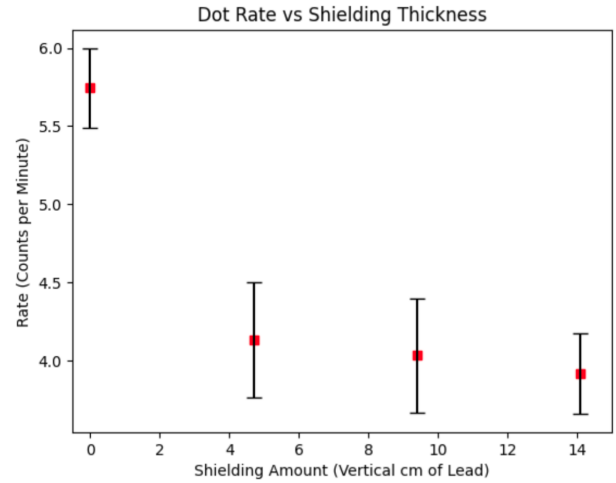


Figure 3: Rates of Dot Detection versus Shielding Thickness

As Figure 2 shows, the rate of dots greatly decreased when the first lead block was added, and did not significantly decrease further afterward. Presuming photons have a short interaction length in lead, this is reasonable, and indicates most were attenuated past the first lead block. The remaining background,

Table 1: Raw Data as Identified by MiniPix Detector

Trial Description	Alphas	Electrons	Muons	Dots	Time (min)
Baseline	7	41	9	397	10
Side Shielding	111	257	53	517	90
1 Vertical Block	19	88	19	124	30
2 Vertical Blocks	24	60	13	121	30
3 Vertical Blocks	54	132	27	235	60

at a rate of about 4 particles per minute, could be attributed to other high-energy particles not easily stopped by lead or easily identified by the MiniPIX.

A similar analysis on particles that the MiniPIX identified as electrons was also performed. The Bethe-Bloch equation applies to electrons, as they are charged, and as such, the results were expected to be highly sensitive to the energy spectrum. Figure 3 gives the observed electron rates and uncertainties.

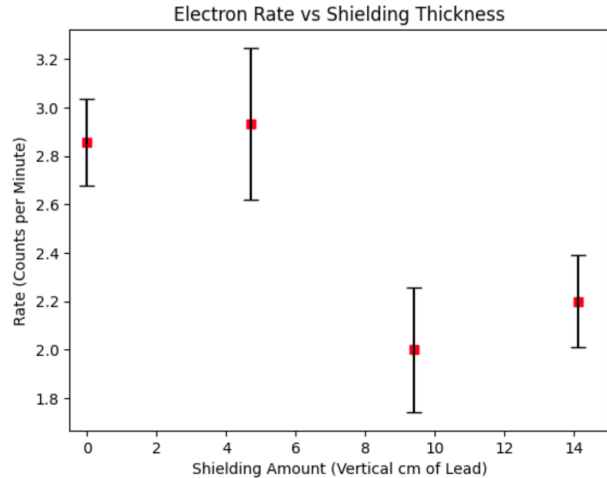


Figure 4: Electron Rate versus Shielding Thickness

As shown in Figure 3, a significant drop in electron counts when the second lead block was added was observed. This might indicate a peak in the electron spectrum, where electrons with energies stoppable by 4.7-9.4 centimeters of lead are more common. The undetected significance of the first block could also be explained by electron showers of high electrons counteracting the shielding effect. For example for a 30GeV electron we expect it to be scattered to order of 100 electrons by the first brick but to be essentially dissipated by the second brick. However, it's also possible that this effect could be attributed to systematic experimental error, which is explored in more detail in the next section. It is also notable that the rates of electrons and muons were less than those of the dots: this might mean that there were

many more high-energy, unidentifiable particles than low-energy, identifiable ones.

4 Additional Considerations & Future Work

To ensure data quality, many potential sources of influence were controlled to the best of our ability. The high location at the top of the observatory minimized the shielding effect of the atmosphere. Additionally, the MiniPIX stayed in a fixed position throughout the experiment and two blocks of lead shielded the sides, isolating only vertical rays and eliminating concerns of shadowing. The temperature of the MiniPIX, which would change the semiconductor detector's ionizing properties, was kept constant over each run through the constant conduction of heat away through the surrounding lead bricks. It is however possible that changes in the temperature of the room, especially as night fell, could have caused a slight difference between runs.

The time of day effects were not appropriately accounted for, though previous experimentation shows the effect of time of day to be 1% fluctuations [1]. In the interest of collecting more data, each trial had an extended length of 30 minutes, causing a large range in time between the first and last data collection trials. Despite the previous results of minimal fluctuation due to time of day, the particle counts from the baseline tests with no shielding varied significantly from the initial trial and the final trial.

The parameters of the experiment changed frequently until the experiment was ultimately conducted, which resulted in uncertainty in methodology. For example, the prevalence of lead bricks was only discovered upon arrival at Jungfrauoch. Additionally, the required relocation to the library eliminated the feasibility of investigating the shielding properties of water. The decision to pivot to only investigating lead-based shielding was one made right before experimental data was collected. The greatest improvement to this experiment would simply be greater

foresight into the parameters and facilities available, which could have been avoided through better communication from the group with the Jungfraujoeh Observatory.

Further testing might involve testing with thinner bricks, allowing for more granularity in thickness and more data points for the curve, testing across a longer stretch of time, like several days, better controlling for any time or day-dependent variables, or testing even higher up to exclusively isolate the effects of lead shielding.

5 Acknowledgments

This experiment would not have been possible without the help of others; the team would like to take this time to sincerely thank the CERN affiliates who gave us the MiniPIX, Daniela and Erich, the caretakers at Jungfraujoeh who allowed us to work with the lead bricks and conduct our measurements, and our professors, Dr. Flanagan and Dr. Lang, for providing invaluable feedback and insight. Thank you, all!

References

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